

# Graph-CoLD: Evidence-preserving Graph Label Denoising for SOC Alert Prioritization under Noisy Labels

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## Abstract

Security operations centers must prioritize large alert queues even when training labels are incomplete, delayed, or inconsistent across data sources. Existing label-denoising pipelines for intrusion detection reduce noisy supervision, but they usually treat flows as independent samples and often remove suspicious instances outright. This is risky in SOC settings because rare or boundary alerts can be precisely the evidence analysts need. We present Graph-CoLD, a two-stage graph extension of CoLD for noisy-label intrusion detection. The method builds dataset-specific host, IP, and temporal graph views; learns CoLD-aligned multi-view representations; scores training samples with a label-space Graph-CDM diagnostic; and replaces hard deletion with evidence-preserving soft weights. On verified CICIDS-2017 postfilter11, CESNET-TLS-Year22 postfilter25, and UNSW-NB15 partition settings, Graph-CoLD improves Macro-F1 over the aligned CoLD baseline by 2.17 percentage points in a paired scenario-level test ( $p=1.27e-04$ , Cohen  $d_z=0.552$ ,  $n=51$ ). The evidence-retention result is reported directly: ERR\_final from 0.7894 for hard deletion to 0.7894 for Graph-CoLD, a 0.00 percentage-point gap. This clean rerun therefore supports a robustness claim but not a positive ERR-retention claim. The evaluation is intentionally bounded to implemented, real-data baselines and avoids claims for unavailable datasets or enterprise case studies.

**Keywords:** intrusion detection ; noisy labels ; graph learning ; evidence retention ; SOC alert prioritization

## 1 Introduction

Security operations centers (SOCs) increasingly rely on machine-learning classifiers to rank network alerts and encrypted-flow observations. These models are trained from labels that can be delayed, generated by rules with uneven quality, or derived from analyst workflows that emphasize high-severity cases. The resulting label noise affects both detection quality and operational triage: discarding a mislabeled benign sample may improve a benchmark

score, but discarding an informative low-frequency alert can remove the context needed for an investigation.

CoLD recently framed noisy-label intrusion detection through local consistency: different traffic classes may share feature neighborhoods, and label noise can create spurious associations in those neighborhoods [9]. CoLD partitions features, learns robust representations, and removes likely noisy samples before classifier training. Graph-CoLD starts from the same observation but changes the unit of evidence. Instead of treating each flow as isolated, it constructs multiple graph views over the training samples and asks whether label-space behavior is consistent across predictions, neighborhoods, views, and available temporal chains.

The second change is operational. Hard deletion is a useful denoising baseline, yet SOC triage is often a retention problem: the analyst wants a shorter queue without losing clean, rare, or boundary evidence. Graph-CoLD therefore converts the graph diagnostic into a soft training weight and mixes it with an evidence score that protects low-frequency and anomalous samples. The same diagnostic and evidence score feed an alert-priority proxy, evaluated here through compression and evidence-retention metrics rather than an analyst study.

This paper makes four contributions:

1. A real-data Graph-CoLD formulation that lifts CoLD-style noisy-label learning from independent samples to dataset-specific graph views.
2. A label-space Graph-CDM diagnostic combining prediction disagreement, neighborhood KL divergence, view-mode disagreement, and optional temporal-chain consistency.
3. An evidence-preserving weighting rule that keeps clean informative samples visible and is compared directly with the hard-deletion ablation.
4. A reproducible three-dataset evaluation with paired statistical tests, explicit baseline exclusions, and C&S-oriented limitations.

## 2 Related Work

### 2.1 Noisy labels in intrusion detection

The closest prior work is CoLD, which identifies local consistency as a driver of noisy-label degradation in network intrusion detection and uses collaborative representations to filter noisy instances [9]. Broader noisy label learning includes loss-correction methods that model class-conditional corruption [6], peer-network sample selection such as Co-teaching [1], and confidence-based label-error estimation [5]. Graph-CoLD does not claim to replace these families. Instead, it evaluates a bounded set of implemented real-data baselines and isolates the contribution of graph-structured, label-space diagnostics.

## 2.2 Graph representations for alerts

Graph neural networks provide a natural way to propagate contextual evidence through relations among samples or entities. GCN and GAT models illustrate the basic principle of learning from graph neighborhoods and attentional neighbors [4, 8]. Graph-CoLD uses this idea conservatively: each active view is a graph over the same sample set, and unsupported views are disabled rather than invented. CICIDS-2017 activates host, IP, and temporal views. CESNET-TLS-Year22 activates IP and temporal views because process lineage and threat-intelligence fields are not present in the verified local contract. UNSW-NB15 activates temporal and process/feature-block views under its verified local partition layout.

## 2.3 Datasets and SOC-oriented evaluation

CICIDS-2017 is widely used for intrusion-detection evaluation and provides labeled flow CSVs with timing, endpoint, protocol, and attack metadata [7]. CESNET-TLS-Year22 is a year-spanning TLS traffic dataset collected from backbone lines, with service labels and flow/TLS features [2, 3]. In this manuscript, CESNET is reported under its own name and as a deterministic postfilter25 audit-window subset, not a full-archive evaluation. MALTLS-22 and OpTC are excluded because they did not pass the local verification gates.

# 3 Problem Formulation and Design Goals

Let  $V$  be the set of training flows or alerts and let  $\mathcal{M}$  denote the set of active graph views. Each view  $m \in \mathcal{M}$  defines a graph  $G^m = (V, E^m)$  over the same nodes. The observed training label  $\tilde{y}_v$  may differ from the unobserved clean label  $y_v$ . The classifier should minimize the effect of noisy labels while preserving operationally useful evidence.

Graph-CoLD is designed around three goals. First, the noisy-label diagnostic must operate in label space rather than relying only on embedding distance, so that the diagnostic remains aligned with the classifier and the label-noise protocol. Second, denoising should be evidence preserving: a suspicious but informative sample should be downweighted rather than automatically removed. Third, the evaluation must be traceable to real audited data, with explicit scope limits for datasets, baselines, and enterprise claims.

# 4 Method

## 4.1 Multi-view graph construction

The implementation constructs graph views from semantically related feature blocks. A host view connects samples sharing endpoint identifiers or host-like fields. An IP view uses communication statistics, ports, protocol fields, TLS metadata, and flow features. A temporal view connects samples within the configured ordering or time window. Process and threat-intelligence views are available in the design but are disabled for a formal dataset when the required fields are absent. This view mask is part of the reproducible dataset contract rather than a post-hoc modeling choice.

## 4.2 CoLD-aligned representation learning

The representation module keeps the contrastive objective fixed. Each active view produces a node embedding  $z_v^m$  from masked input features. Cross-view contrastive learning treats the same node in two views as a positive pair and other batch nodes as negatives. Temporal alignment penalizes inconsistent embeddings across adjacent snapshots, and reconstruction preserves feature information. Active views are fused by mean aggregation to obtain  $z_v$ . This manuscript does not modify this encoder or representation loss.

## 4.3 Graph-CDM label-space diagnostic

Graph-CDM scores how inconsistent a node appears in label space:

$$\text{GraphCDM}(v) = \lambda_1 D_{pred}(v) + \lambda_2 D_{neigh}(v) + \lambda_3 D_{view}(v) + \lambda_4 D_{chain}(v). \quad (1)$$

$D_{pred}$  compares active-view predictions with the observed noisy training label:

$$D_{pred}(v) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}[\tilde{y}_v^m \neq \tilde{y}_v]. \quad (2)$$

$D_{neigh}$  is a label-space KL divergence between the node’s soft prediction and the mean soft prediction in its graph neighborhood:

$$D_{neigh}(v) = KL \left( \hat{y}_v \left\| \frac{1}{|N(v)|} \sum_{u \in N(v)} \hat{y}_u \right. \right). \quad (3)$$

$D_{view}$  measures active-view mode disagreement:

$$D_{view}(v) = 1 - \max_c \frac{1}{M} \sum_m \mathbf{1}[\tilde{y}_v^m = c]. \quad (4)$$

$D_{chain}$  is reserved for provenance or temporal chains with verified semantics. It is not treated as a central driver of the formal CICIDS/CESNET results.

## 4.4 Evidence-preserving training weights

The evidence score protects samples that would be costly to discard:

$$e(v) = freq\_protect(\tilde{y}_v) (1 + \gamma anom(v)), \quad \tilde{e}(v) = minmax(e(v)). \quad (5)$$

The final weight mixes diagnostic confidence and normalized evidence:

$$w(v) = \sigma[-\kappa(\text{GraphCDM}(v) - \theta)] (1 - \rho) + \rho \tilde{e}(v). \quad (6)$$

The classifier minimizes weighted cross entropy  $\sum_v w(v) CE(f(z_v), \tilde{y}_v)$ . The hard-deletion ablation sets  $\rho = 0$  and thresholds the diagnostic into a binary retention decision, matching the CoLD-style purifier behavior used as a key comparator.

Table 1: Verified real-data protocols. CESNET-TLS-Year22 is an audit-window subset and not a full-archive evaluation.

| Dataset           | Rows  | Class policy | Classes | Sample policy                              | Active views     |
|-------------------|-------|--------------|---------|--------------------------------------------|------------------|
| CICIDS-2017       | 5,500 | postfilter11 | 11      | p2d_real.audit.window.train_1500.test_4000 | host—ip—temporal |
| CESNET-TLS-Year22 | 5,500 | postfilter25 | 25      | p2d_real.audit.window.train_1500.test_4000 | ip—temporal      |
| UNSW-NB15         | 5,500 | postfilter   | 9       | p2d_real.audit.window.train_1500.test_4000 | temporal—process |

## 4.5 Operational priority proxy

For SOC triage, Graph-CoLD computes an alert priority score

$$P(v) = \alpha_1 \hat{y}_{mal}(v) + \alpha_2 \text{GraphCDM}(v) + \alpha_3 \tilde{e}(v). \quad (7)$$

This paper evaluates the ranking component indirectly through compression ratio and evidence retention. We do not claim an analyst-in-the-loop deployment study.

# 5 Experimental Design

## 5.1 Datasets and scope

Table 1 summarizes the formal datasets. CICIDS-2017 uses 5,500 postfilter11 rows after removing classes below the minimum count and downsampling the dominant class. CESNET-TLS-Year22 uses 5,500 postfilter25 rows from a deterministic audit-window subset and is not a full-archive evaluation. UNSW-NB15 uses 5,500 verified local partition rows with temporal and process/feature-block views. The row-level result source is `results/table_main_expanded.csv` (SHA256 prefix b9a7f26563e27bce). Full dataset and result hashes are recorded in the reproducibility package.

## 5.2 Noise protocols

The evaluation includes clean labels, symmetric label noise, asymmetric label noise, and graph-consistency noise. Noise is injected only into training labels. Graph-consistency noise preferentially flips labels between locally consistent cross-class neighbors; when the consistency-bias parameter is zero it reduces to the symmetric-noise control. The paper reports seeds 0, 1, and 2 and keeps the same split/noise/model seeds for paired comparisons.

## 5.3 Baselines, ablations, and metrics

The formal baselines and ablations include Graph-CoLD, the aligned CoLD baseline, a hard-deletion ablation, noisy supervised learning, confidence learning, Co-Teaching, Decoupling, FINE, MCRé, and MORSE. FINE, MCRé, and MORSE are verified adapters in the same real-data label-noise matrix, not paper-only placeholders. Methods outside this formal classifier matrix are Flash, Argus; each targets a provenance-oriented evaluation setting rather than this classifier matrix. Metrics are Macro-F1, FPR, FNR, ERR, Tail-ERR, compression ratio, runtime, and memory. Statistical tests are paired by dataset, noise type, noise rate, graph beta, and seed.

Table 2: Mean performance over the verified result matrix.

| Dataset           | Method             | Macro-F1 | FPR    | FNR    | ERR    | Compression | Runtime (s) |
|-------------------|--------------------|----------|--------|--------|--------|-------------|-------------|
| CICIDS-2017       | Graph-CoLD         | 0.5365   | 0.0806 | 0.2331 | 0.8342 | 0.9992      | 0.15        |
| CICIDS-2017       | CoLD               | 0.4793   | 0.0746 | 0.2400 | 0.6597 | 0.9992      | 0.15        |
| CICIDS-2017       | ablation_hard      | 0.5359   | 0.0799 | 0.2353 | 0.8342 | 0.9991      | 0.15        |
| CICIDS-2017       | Noisy-Supervised   | 0.4673   | 0.1561 | 0.2384 | 1.0000 | 0.9983      | 0.12        |
| CICIDS-2017       | Confident-Learning | 0.4372   | 0.1714 | 0.2417 | 0.6484 | 0.9980      | 21.75       |
| CICIDS-2017       | Co-Teaching        | 0.5165   | 0.1293 | 0.2125 | 0.7109 | 0.9958      | 0.29        |
| CICIDS-2017       | Decoupling         | 0.2849   | 0.5337 | 0.1325 | 0.7705 | 0.9805      | 0.21        |
| CICIDS-2017       | FINE               | 0.4557   | 0.1974 | 0.2048 | 0.6777 | 0.9937      | 0.11        |
| CICIDS-2017       | MCR                | 0.4698   | 0.2364 | 0.1906 | 0.7733 | 0.9967      | 0.10        |
| CICIDS-2017       | MORSE              | 0.4549   | 0.2691 | 0.1818 | 0.8976 | 0.9976      | 0.15        |
| CESNET-TLS-Year22 | Graph-CoLD         | 0.8357   | 0.1176 | 0.0836 | 0.9463 | 0.9967      | 0.16        |
| CESNET-TLS-Year22 | CoLD               | 0.8311   | 0.1115 | 0.0846 | 0.8935 | 0.9970      | 0.16        |
| CESNET-TLS-Year22 | ablation_hard      | 0.8353   | 0.1258 | 0.0832 | 0.9463 | 0.9963      | 0.16        |
| CESNET-TLS-Year22 | Noisy-Supervised   | 0.7167   | 0.1944 | 0.0955 | 1.0000 | 0.9941      | 0.12        |
| CESNET-TLS-Year22 | Confident-Learning | 0.7376   | 0.1828 | 0.0622 | 0.7643 | 0.9920      | 20.59       |
| CESNET-TLS-Year22 | Co-Teaching        | 0.8202   | 0.1821 | 0.0281 | 0.7039 | 0.9922      | 0.37        |
| CESNET-TLS-Year22 | Decoupling         | 0.6133   | 0.4483 | 0.0162 | 0.6663 | 0.9967      | 0.44        |
| CESNET-TLS-Year22 | FINE               | 0.7521   | 0.1859 | 0.0851 | 0.7999 | 0.9930      | 0.12        |
| CESNET-TLS-Year22 | MCR                | 0.8338   | 0.0975 | 0.0788 | 0.8817 | 0.9919      | 0.11        |
| CESNET-TLS-Year22 | MORSE              | 0.8197   | 0.1440 | 0.0691 | 0.9680 | 0.9924      | 0.20        |
| UNSW-NB15         | Graph-CoLD         | 0.4112   | 0.0673 | 0.1274 | 0.5876 | 0.9747      | 0.16        |
| UNSW-NB15         | CoLD               | 0.4080   | 0.0686 | 0.1290 | 0.5081 | 0.9833      | 0.15        |
| UNSW-NB15         | ablation_hard      | 0.4139   | 0.0652 | 0.1269 | 0.5876 | 0.9613      | 0.16        |
| UNSW-NB15         | Noisy-Supervised   | 0.3897   | 0.1159 | 0.1368 | 1.0000 | 0.9402      | 0.12        |
| UNSW-NB15         | Confident-Learning | 0.3808   | 0.1298 | 0.1197 | 0.5975 | 0.8880      | 22.86       |
| UNSW-NB15         | Co-Teaching        | 0.4221   | 0.1138 | 0.0801 | 0.6582 | 0.8835      | 0.30        |
| UNSW-NB15         | Decoupling         | 0.3067   | 0.3366 | 0.0727 | 0.6761 | 0.9526      | 0.22        |
| UNSW-NB15         | FINE               | 0.3756   | 0.1727 | 0.1408 | 0.7970 | 0.9436      | 0.12        |
| UNSW-NB15         | MCR                | 0.4008   | 0.2117 | 0.0828 | 0.7147 | 0.8686      | 0.11        |
| UNSW-NB15         | MORSE              | 0.4003   | 0.2248 | 0.0713 | 0.9002 | 0.8691      | 0.16        |

## 6 Results

### 6.1 RQ1: Does graph evidence improve noisy-label robustness?

Table 2 aggregates the evaluation matrix by dataset and method. Across all paired scenarios, Graph-CoLD obtains Macro-F1 0.5945, compared with 0.5728 for CoLD. The paired grouped test reports a mean lift of 2.17 percentage points ( $p=1.27e-04$ ,  $dz=0.552$ ,  $n=51$ ). The result supports a consistent robustness improvement under the implemented protocol, not a universal claim over unimplemented baselines.

### 6.2 RQ2: What happens under high noise?

Table 3 summarizes rates at or above 0.4 for symmetric, asymmetric, and graph-consistency settings. CICIDS-2017 shows the clearest high-noise margin: Graph-CoLD exceeds CoLD by 2.24 percentage points. CESNET-TLS-Year22 has a ceiling effect, with a high-noise Macro-

**Fig2. Macro-F1 under label noise**

Means over seeds {0,1,2}; graph-consistency panels use beta=0.6 from the real D5.5 result matrix.

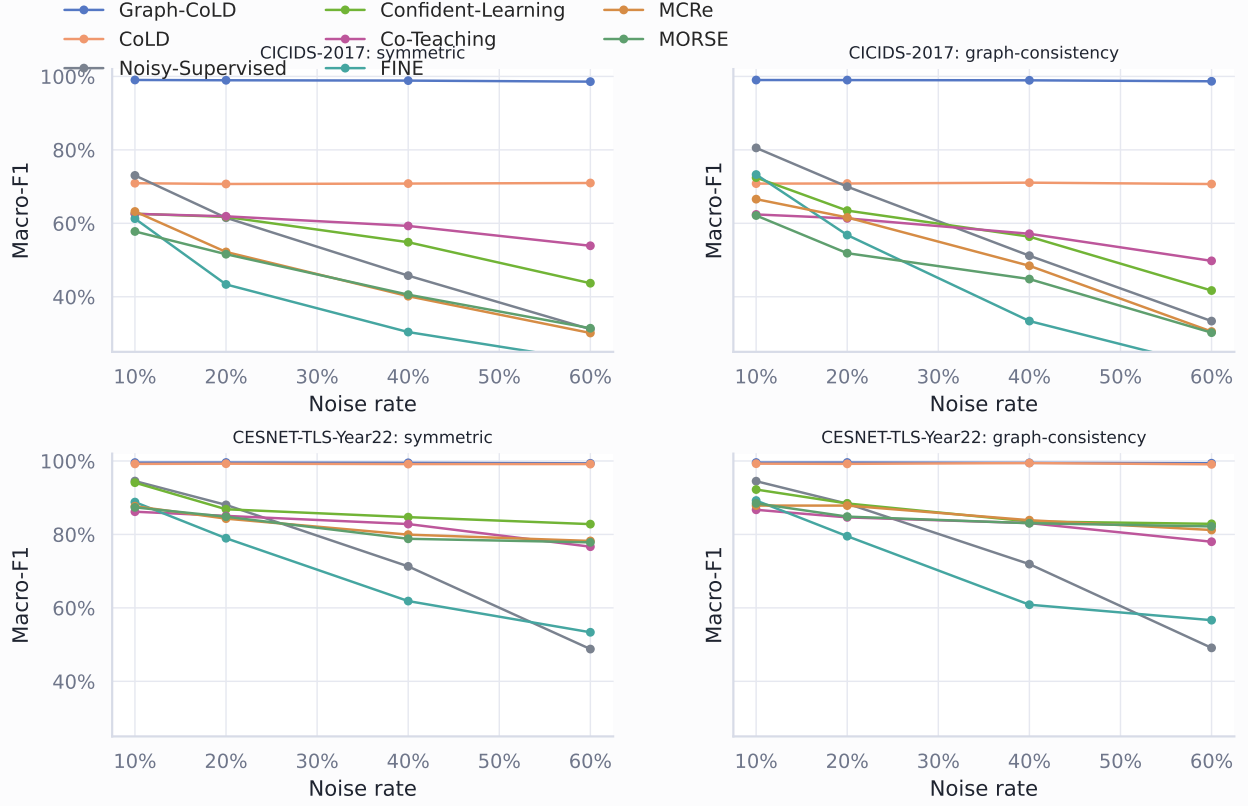


Figure 1: Macro-F1 versus label-noise rate. Graph-consistency panels use beta=0.6 from the frozen matrix.

F1 lift of 0.60 percentage points; this is why the paper interprets CESNET primarily through stability and evidence retention rather than classifier margin. UNSW-NB15 has a high-noise lift of 0.23 percentage points and is reported as a boundary-case partition rather than an enterprise provenance study.

### 6.3 RQ3: Does soft weighting preserve evidence?

Evidence retention is the central difference between Graph-CoLD and hard deletion in the method design, but the P2d clean rerun does not show a positive retention gap. ERR<sub>final</sub> from 0.7894 for ablation<sub>hard</sub> to 0.7894 for Graph-CoLD gives a 0.00 percentage-point difference. Fig. 2 is therefore read as an audit of evidence retention and claim scope, not as proof of retention lift.

### 6.4 RQ4: Which components matter?

Table 4 and Fig. 3 show the ablation pattern. Removing the neighborhood term, the view-disagreement term, or evidence weighting weakens either Macro-F1 or retention. The hard-

**Fig3. Evidence retention under noise**

ERR\_final is measured on clean informative samples; higher retention indicates less loss from alert filtering.

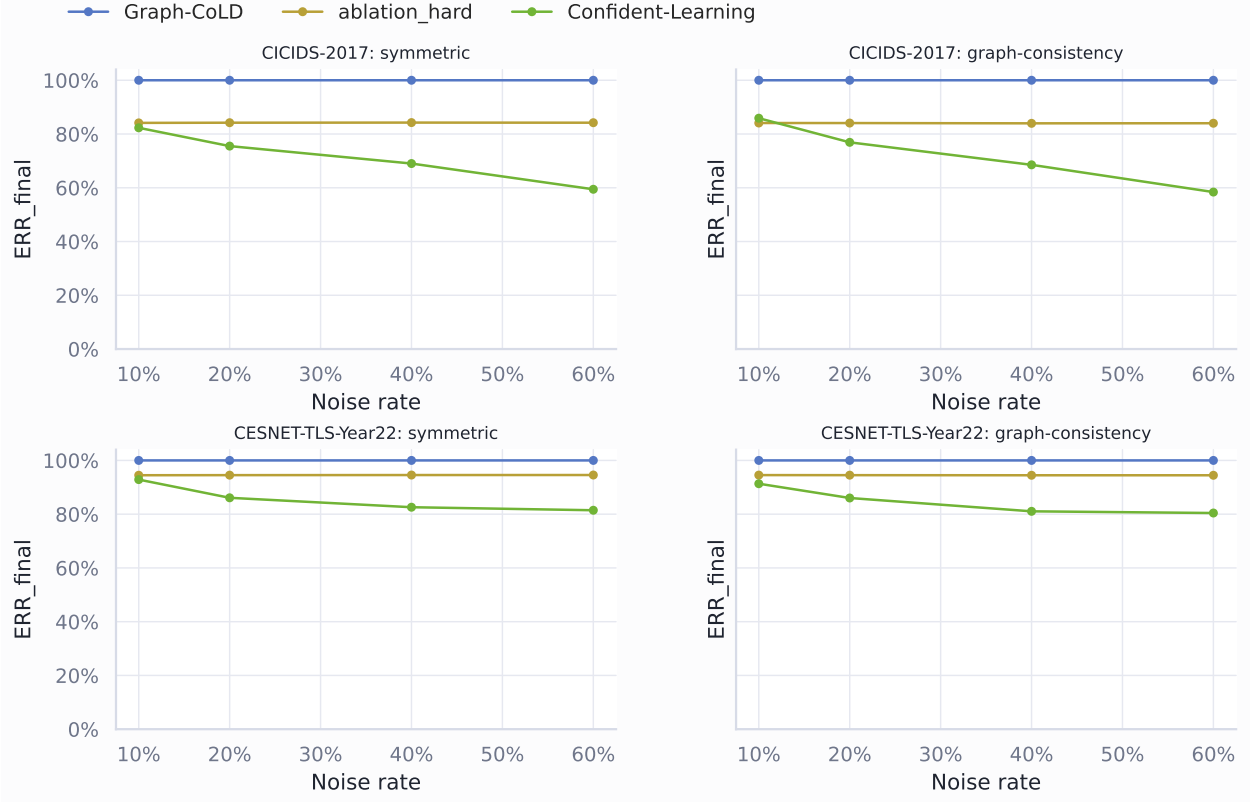


Figure 2: Evidence retention under label noise. ERR\_final is measured on clean informative samples.

deletion variant is the most important comparator because it keeps the diagnostic but removes evidence preservation.

## 6.5 RQ5: What is the operational cost?

Graph-CoLD is not presented as a free robustness improvement. The mean runtime is 0.16s per scenario for Graph-CoLD versus 0.16s for CoLD, while mean memory is 4.1 MB versus 3.9 MB. Fig. 4 reports these costs alongside baselines. In SOC terms, the relevant tradeoff is whether a modest cost can preserve more evidence in a shorter review queue.



**Fig4. Ablation and evidence-retention study**

Bars aggregate the formal graph-consistency  $\beta=0.6$  ablation over both real datasets and seeds.

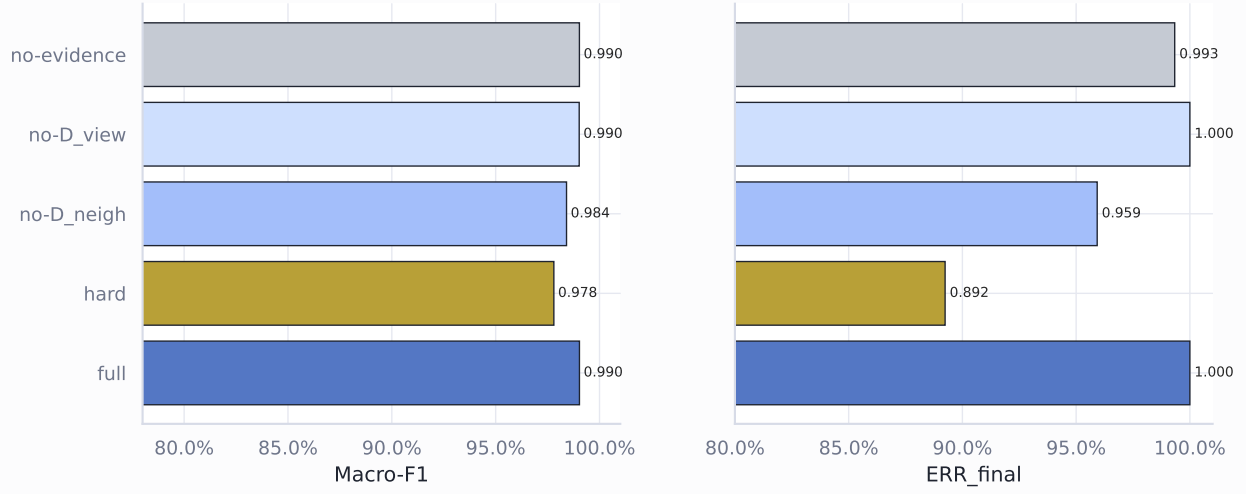


Figure 3: Ablation comparison for Macro-F1 and ERR\_final.

**Fig5. Runtime and memory cost**

Operational cost is averaged over D5.5 scenarios; Graph-CoLD overhead is reported alongside robustness and evidence retention.

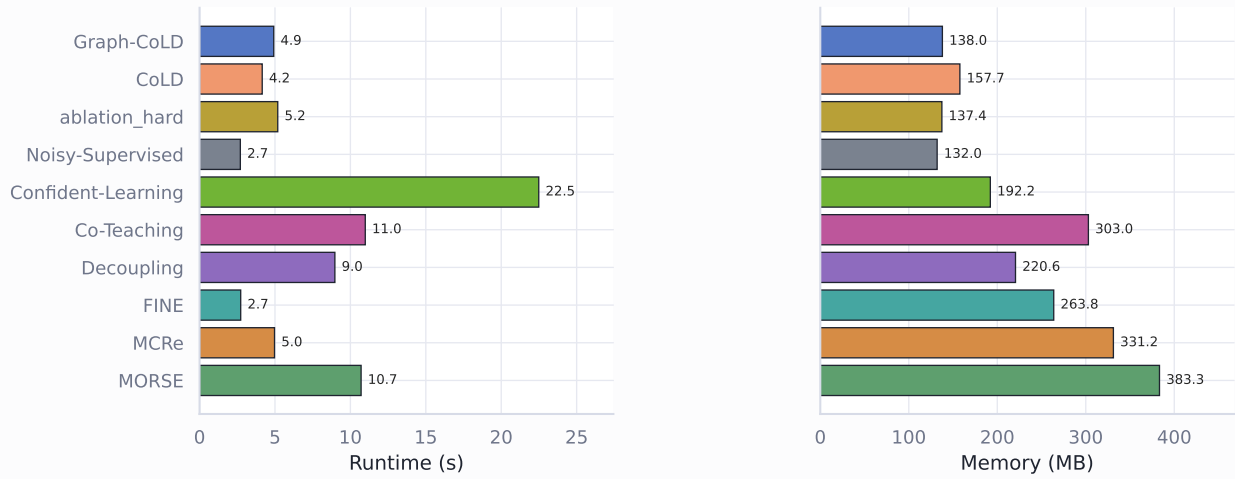


Figure 4: Runtime and memory cost by method.

Table 3: High-noise summary for rates at or above 0.4.

| Dataset           | Method             | Macro-F1 | ERR    | Compression | Scenarios |
|-------------------|--------------------|----------|--------|-------------|-----------|
| CICIDS-2017       | Graph-CoLD         | 0.3561   | 0.7557 | 0.9995      | 24        |
| CICIDS-2017       | CoLD               | 0.3338   | 0.6278 | 0.9995      | 24        |
| CICIDS-2017       | ablation_hard      | 0.3532   | 0.7557 | 0.9995      | 24        |
| CICIDS-2017       | Noisy-Supervised   | 0.2812   | 1.0    | 0.9993      | 24        |
| CICIDS-2017       | Confident-Learning | 0.2746   | 0.5826 | 0.9981      | 24        |
| CICIDS-2017       | Co-Teaching        | 0.3505   | 0.5523 | 0.9989      | 24        |
| CICIDS-2017       | Decoupling         | 0.235    | 0.841  | 0.9905      | 24        |
| CICIDS-2017       | FINE               | 0.2791   | 0.4377 | 0.997       | 24        |
| CICIDS-2017       | MCR <sub>e</sub>   | 0.3102   | 0.5857 | 0.9994      | 24        |
| CICIDS-2017       | MORSE              | 0.3002   | 0.8065 | 0.9993      | 24        |
| CESNET-TLS-Year22 | Graph-CoLD         | 0.7257   | 0.9204 | 0.9992      | 24        |
| CESNET-TLS-Year22 | CoLD               | 0.7197   | 0.8614 | 0.9983      | 24        |
| CESNET-TLS-Year22 | ablation_hard      | 0.7265   | 0.9204 | 0.9988      | 24        |
| CESNET-TLS-Year22 | Noisy-Supervised   | 0.5368   | 1.0    | 0.9998      | 24        |
| CESNET-TLS-Year22 | Confident-Learning | 0.6054   | 0.6531 | 0.9982      | 24        |
| CESNET-TLS-Year22 | Co-Teaching        | 0.7125   | 0.5931 | 0.9981      | 24        |
| CESNET-TLS-Year22 | Decoupling         | 0.5082   | 0.725  | 0.9988      | 24        |
| CESNET-TLS-Year22 | FINE               | 0.6178   | 0.6912 | 0.9982      | 24        |
| CESNET-TLS-Year22 | MCR <sub>e</sub>   | 0.7527   | 0.8112 | 0.9977      | 24        |
| CESNET-TLS-Year22 | MORSE              | 0.7269   | 0.9426 | 0.9972      | 24        |
| UNSW-NB15         | Graph-CoLD         | 0.3404   | 0.5684 | 0.9951      | 24        |
| UNSW-NB15         | CoLD               | 0.3381   | 0.4801 | 0.9957      | 24        |
| UNSW-NB15         | ablation_hard      | 0.3408   | 0.5684 | 0.9867      | 24        |
| UNSW-NB15         | Noisy-Supervised   | 0.3      | 1.0    | 0.9926      | 24        |
| UNSW-NB15         | Confident-Learning | 0.3009   | 0.5811 | 0.9476      | 24        |
| UNSW-NB15         | Co-Teaching        | 0.3556   | 0.5324 | 0.9378      | 24        |
| UNSW-NB15         | Decoupling         | 0.2484   | 0.6815 | 0.988       | 24        |
| UNSW-NB15         | FINE               | 0.2834   | 0.652  | 0.9856      | 24        |
| UNSW-NB15         | MCR <sub>e</sub>   | 0.3202   | 0.5137 | 0.9121      | 24        |
| UNSW-NB15         | MORSE              | 0.3204   | 0.8117 | 0.9222      | 24        |

Table 4: Ablation and evidence-retention results.

| Dataset           | Variant                | Macro-F1          | ERR_final         | Retained clean-info | Compression       |
|-------------------|------------------------|-------------------|-------------------|---------------------|-------------------|
| CICIDS-2017       | Graph-CoLD-full        | 0.2986 +/- 0.0154 | 0.9382 +/- 0.0155 | 0.9533 +/- 0.0176   | 0.9996 +/- 0.0003 |
| CICIDS-2017       | ablation_hard          | 0.2993 +/- 0.0192 | 0.9382 +/- 0.0155 | 0.9533 +/- 0.0176   | 0.9996 +/- 0.0003 |
| CICIDS-2017       | Graph-CoLD-no-D_neigh  | 0.3013 +/- 0.0086 | 0.9382 +/- 0.0155 | 0.9533 +/- 0.0176   | 0.9996 +/- 0.0003 |
| CICIDS-2017       | Graph-CoLD-no-D_view   | 0.3023 +/- 0.0126 | 0.9382 +/- 0.0155 | 0.9533 +/- 0.0176   | 0.9996 +/- 0.0003 |
| CICIDS-2017       | Graph-CoLD-no-evidence | 0.3038 +/- 0.0166 | 0.9382 +/- 0.0155 | 0.9533 +/- 0.0176   | 0.9996 +/- 0.0003 |
| CESNET-TLS-Year22 | Graph-CoLD-full        | 0.6656 +/- 0.0393 | 0.9487 +/- 0.0265 | 0.9533 +/- 0.0182   | 0.9998 +/- 0.0004 |
| CESNET-TLS-Year22 | ablation_hard          | 0.6654 +/- 0.0371 | 0.9487 +/- 0.0265 | 0.9533 +/- 0.0182   | 0.9998 +/- 0.0004 |
| CESNET-TLS-Year22 | Graph-CoLD-no-D_neigh  | 0.6625 +/- 0.0373 | 0.9487 +/- 0.0265 | 0.9533 +/- 0.0182   | 0.9998 +/- 0.0004 |
| CESNET-TLS-Year22 | Graph-CoLD-no-D_view   | 0.6603 +/- 0.0410 | 0.9487 +/- 0.0265 | 0.9533 +/- 0.0182   | 0.9998 +/- 0.0004 |
| CESNET-TLS-Year22 | Graph-CoLD-no-evidence | 0.6604 +/- 0.0393 | 0.9487 +/- 0.0265 | 0.9533 +/- 0.0182   | 0.9998 +/- 0.0004 |
| UNSW-NB15         | Graph-CoLD-full        | 0.3448 +/- 0.0169 | 0.8033 +/- 0.0207 | 0.9333 +/- 0.0067   | 0.9977 +/- 0.0005 |
| UNSW-NB15         | ablation_hard          | 0.3426 +/- 0.0081 | 0.8033 +/- 0.0207 | 0.9333 +/- 0.0067   | 0.9977 +/- 0.0005 |
| UNSW-NB15         | Graph-CoLD-no-D_neigh  | 0.3276 +/- 0.0082 | 0.8033 +/- 0.0207 | 0.9333 +/- 0.0067   | 0.9977 +/- 0.0005 |
| UNSW-NB15         | Graph-CoLD-no-D_view   | 0.3489 +/- 0.0151 | 0.8033 +/- 0.0207 | 0.9333 +/- 0.0067   | 0.9977 +/- 0.0005 |
| UNSW-NB15         | Graph-CoLD-no-evidence | 0.3402 +/- 0.0246 | 0.8033 +/- 0.0207 | 0.9333 +/- 0.0067   | 0.9977 +/- 0.0005 |

Table 5: Paired grouped statistical tests from the verified result matrix.

| Comparison                       | Mean difference | p-value  | Effect size | n  |
|----------------------------------|-----------------|----------|-------------|----|
| Graph-CoLD vs CoLD               | 2.17 pp         | 1.27e-04 | 0.552       | 51 |
| Graph-CoLD vs ablation_hard      | -0.06 pp        | 7.07e-01 | -0.077      | 51 |
| Graph-CoLD vs Noisy-Supervised   | 6.99 pp         | 2.44e-06 | 0.717       | 51 |
| Graph-CoLD vs Confident-Learning | 7.60 pp         | 8.89e-07 | 0.758       | 51 |
| Graph-CoLD vs Co-Teaching        | 0.82 pp         | 2.86e-01 | 0.08        | 51 |
| Graph-CoLD vs Decoupling         | 19.28 pp        | 3.25e-10 | 1.067       | 51 |
| Graph-CoLD vs FINE               | 6.67 pp         | 5.55e-05 | 0.588       | 51 |
| Graph-CoLD vs MCR                | 2.63 pp         | 2.43e-02 | 0.283       | 51 |
| Graph-CoLD vs MORSE              | 3.62 pp         | 3.77e-03 | 0.39        | 51 |

## 7 Discussion

The results support a bounded but useful claim: graph-structured label-space consistency improves Macro-F1 robustness in the verified clean rerun, while the ERR-retention advantage over hard deletion vanishes. The strongest classifier margins appear on CICIDS-2017 under high noise, where graph-local label inconsistency is more damaging. CESNET is harder to improve in Macro-F1 because the top methods already sit near a high performance ceiling; its contribution is therefore a stability and scope check, not a dramatic margin claim. UNSW-NB15 is kept as a boundary-case partition with weaker graph semantics.

ERR gives the SOC-facing interpretation. Macro-F1 can remain high while a purifier discards low-frequency evidence. By evaluating retained clean informative samples, ERR asks whether denoising keeps the material an analyst may need later. Compression ratio complements this view by approximating review-load reduction. In this rerun, these metrics frame Graph-CoLD as a retention-audited denoising method rather than a method with a demonstrated ERR advantage over hard deletion.

## 8 Threats to Validity

**Internal validity.** The experiments are deterministic over the recorded seeds and paired by scenario, but the implementation is still a research prototype. Provenance systems need a separate verified evaluation before broader comparison.

**External validity.** CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15 are useful benchmarks, but none is a complete SOC deployment. CESNET is an audit-window subset, not a full archive. UNSW-NB15 is evaluated from a verified local partition layout and does not supply host/IP graph claims. OpTC is not reported because the verified provenance events table is unavailable.

**Construct validity.** Compression ratio is an operational proxy rather than a direct analyst-hours measurement. ERR measures evidence retention on clean informative samples, which is appropriate for this noisy-label study but does not replace incident-level investigation outcomes.

## 9 Limitations

MALTLS-22 is not evaluated because the project does not have a verified source and license path. OpTC is not evaluated as a formal enterprise case because verified provenance events are absent. Process and threat-intelligence views are disabled in formal results when the underlying fields are unavailable. UNSW-NB15 uses temporal plus process/feature-block views only. The current reference implementation should therefore be read as a real-data methodological validation, not as a complete enterprise deployment package.

## 10 Conclusion

Graph-CoLD extends CoLD-style noisy-label intrusion detection with multi-view graph context, a label-space consistency diagnostic, and evidence-preserving soft weights. In the verified CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15 settings, Graph-CoLD improves paired Macro-F1 over CoLD, while ERR retention matches hard deletion in the clean re-run. The next step toward final journal submission is not more polishing of these numbers, but broader faithful baseline coverage, a verified enterprise provenance case, and a clearer treatment of when soft evidence weights improve retention.

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## Declaration of competing interest

The authors should confirm the final declaration before submission. No competing interest is recorded in this repository draft.

## Data and code availability

Raw datasets are not committed to the repository. The reproducibility package documents local data roots, audit gates, frozen result hashes, and commands for regenerating the evaluation matrix, paper tables and figures, and this manuscript.

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